Year 11 Mathematics

Graphs 6

8.3 Non-linear Simultaneous Equations

Non-linear Simultaneous Equations

STARTER · 5 MIN

Date: _____ **Title:** Non-linear Simultaneous Equations

LEARNING OBJECTIVES

- Solve a linear and a quadratic equation using graphs
- Solve a linear equation and a circle using graphs
- Solve non-linear simultaneous equations by substitution

STARTER *Answer in your book. No calculators.*

1. Expand $(x + 2)^2$
2. Solve $x^2 - 5x + 6 = 0$
3. Write down the radius of $x^2 + y^2 = 25$
4. Substitute $y = 2x$ into $x + y = 9$ and solve

Starter — Answers

STARTER

Tick or correct in green pen

1. $x^2 + 4x + 4$

2. $(x-2)(x-3)=0, \quad x=2$ and $x=\mathbf{3}$

3. $\sqrt{25}=\mathbf{5}$

4. $3x=9, \quad x=\mathbf{3}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve a linear and a quadratic equation using graphs	☐
SILVER	I can solve a linear equation and a circle using graphs	☐
GOLD	I can solve non-linear simultaneous equations by substitution	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Non-linear Simultaneous Equations

INTRODUCTION · 3 MIN

A line can cross a curve or circle at two points, so there can be two pairs of answers.

Graphically

Solutions are the crossing points.

Give both x and y

Circle

$x^2 + y^2 = r^2$ has centre (0, 0).

$x^2 + y^2 = 16$: $r = 4$

Substitution

Replace y with the linear expression.

This gives a quadratic

Pairs

Each x has its own y.

Up to two solutions

Examples 1 and 2 — Reading Crossing Points

EXAMPLES · 5 MIN

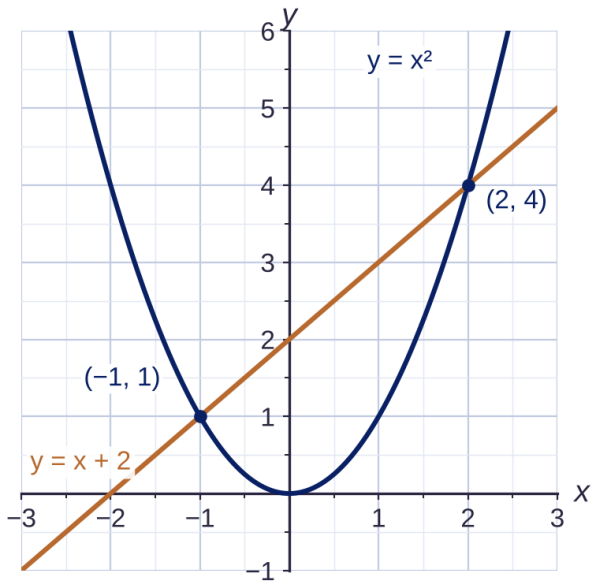
BRONZE

Watch and copy into your book

Hint solutions: the points where the graphs cross

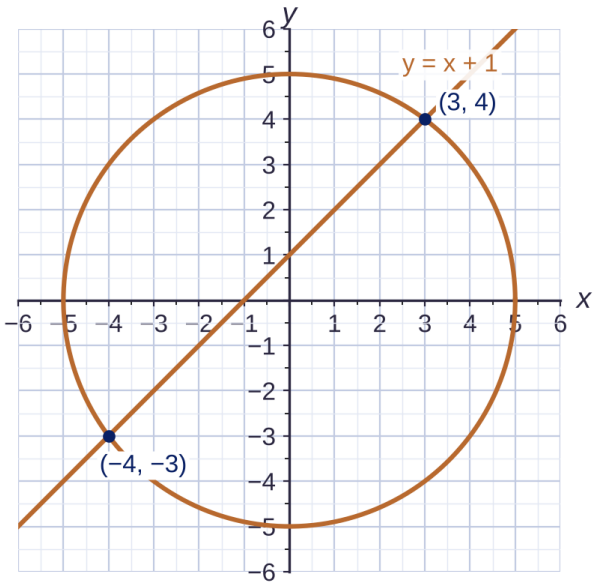
Example 1

Use the graph to solve $y = x^2$ and $y = x + 2$.



Example 2

Use the graph to solve $x^2 + y^2 = 25$ and $y = x + 1$.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Read the crossing points

(−1, 1) and (2, 4)

Tip: give x and y for each point.

Example 2

Read the crossing points

(3, 4) and (−4, −3)

Tip: the circle has radius 5.

Task 1

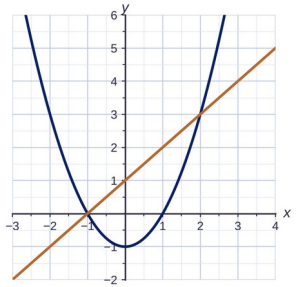
PRACTICE · 9 MIN

BRONZE

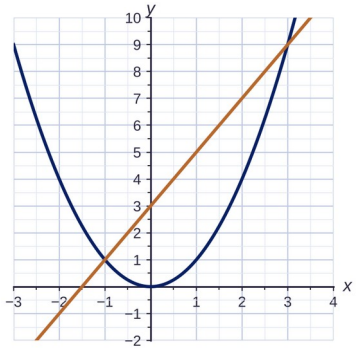
Write down the solutions from each graph

Hint solutions: the points where the graphs cross

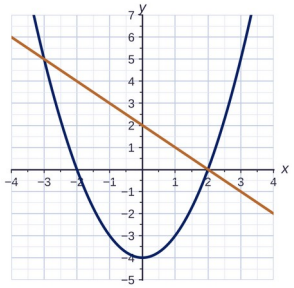
1 $y = x^2 - 1$ and $y = x + 1$



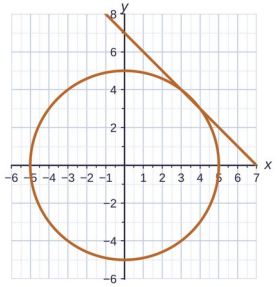
2 $y = x^2$ and $y = 2x + 3$



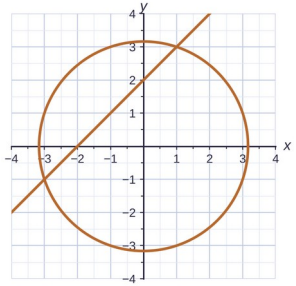
3 $y = x^2 - 4$ and $y = 2 - x$



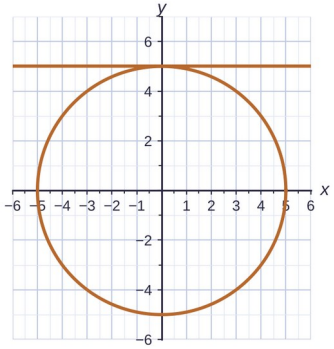
4 $x^2 + y^2 = 25$ and $y = 7 - x$



5 $x^2 + y^2 = 10$ and $y = x + 2$



★ $x^2 + y^2 = 25$ and $y = 5$



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $(-1,\ 0)$ and $(2,\ 3)$
2. $(-1,\ 1)$ and $(3,\ 9)$
3. $(-3,\ 5)$ and $(2,\ 0)$
4. $(3,\ 4)$ and $(4,\ 3)$
5. $(-3,\ -1)$ and $(1,\ 3)$

★ $(0,\ 5)$ only

Examples 3 and 4 — Solving by Substitution

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint substitute the linear equation into the other

Example 3

Solve $y = x^2 - 3x + 4$ and $y = x + 1$.

Example 4

Solve $y = x^2$ and $x + y = 6$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x^2 - 3x + 4 = x + 1$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

(1, 2) and (3, 4)

Tip: find y using the linear equation.

Example 4

$$x + x^2 = 6$$

$$x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0$$

(2, 4) and (−3, 9)

Tip: substitute $y = x^2$ into the line.

Task 2

PRACTICE · 9 MIN

SILVER

Solve each pair algebraically

Hint substitute the linear equation into the other

1. $y = x^2$ and $y = x + 6$

2. $y = x^2 + 2$ and $y = 3x$

3. $y = x^2 - 5$ and $y = 2x + 3$

4. $y = 2x^2$ and $y = x + 1$

5. $y = x^2 - 4x$ and $y = x - 6$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x^2 - x - 6 = 0$, **(3, 9) and (−2, 4)**

2. $x^2 - 3x + 2 = 0$, **(1, 3) and (2, 6)**

3. $x^2 - 2x - 8 = 0$, **(4, 11) and (−2, −1)**

4. $2x^2 - x - 1 = 0$, **(1, 2) and (−0.5, 0.5)**

5. $x^2 - 5x + 6 = 0$, **(2, −4) and (3, −3)**

Examples 5 and 6 — Lines and Circles

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint substitute the linear equation into the other

Example 5

(4 marks)

Solve $x^2 + y^2 = 25$ and $y = x + 1$.

Example 6

(3 marks)

Solve $x^2 + y^2 = 20$ and $y = 2x$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$x^2 + (x + 1)^2 = 25$$

$$2x^2 + 2x - 24 = 0$$

$$(x + 4)(x - 3) = 0$$

(3, 4) and (−4, −3)

Tip: expand the bracket carefully.

Example 6

$$x^2 + (2x)^2 = 20$$

$$5x^2 = 20$$

(2, 4) and (−2, −4)

Tip: $x^2 = 4$ gives $x = \pm 2$.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint substitute the linear equation into the other

1. Solve $x^2 + y^2 = 10$ and $y = x + 2$ (4 marks)

2. Solve $x^2 + y^2 = 25$ and $y = 7 - x$ (4 marks)

3. Solve $x^2 + y^2 = 13$ and $y = x + 1$ (4 marks)

4. Solve $x^2 + y^2 = 50$ and $y = x$ (3 marks)

5. Show that the line $y = x + 10$ does not meet the circle $x^2 + y^2 = 25$ (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $2x^2 + 4x - 6 = 0$, **(1, 3) and (−3, −1)**

2. $2x^2 - 14x + 24 = 0$, **(3, 4) and (4, 3)**

3. $2x^2 + 2x - 12 = 0$, **(2, 3) and (−3, −2)**

4. $2x^2 = 50$, **(5, 5) and (−5, −5)**

5. $2x^2 + 20x + 75 = 0$, $b^2 - 4ac = -200 < 0$, **no solutions**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $y = x^2$ and $y = 9$

SILVER Solve $y = x^2$ and $y = x + 2$

GOLD Solve $x^2 + y^2 = 8$ and $y = x$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x^2 = 9$, **(3, 9) and (−3, 9)**

SILVER $x^2 - x - 2 = 0$, **(2, 4) and (−1, 1)**

GOLD $2x^2 = 8$, **(2, 2) and (−2, −2)**

Exam Practice — Reading Crossing Points

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

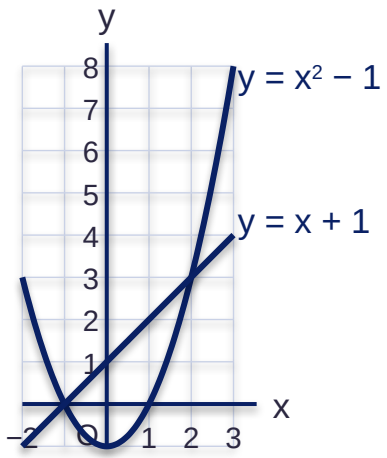
Hint solutions: the points where the graphs cross

Question 1

(2 marks)

$y = x^2 - 1$ and $y = x + 1$ are drawn.

Use the graph to solve the equations.



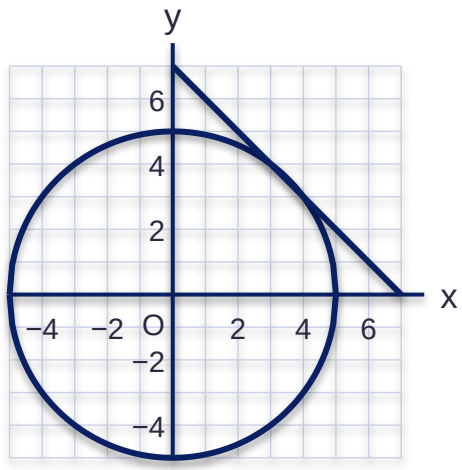
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Question 2

(2 marks)

$x^2 + y^2 = 25$ and $x + y = 7$ are drawn.

Write down the solutions.



.....

Exam Practice — Solving by Substitution

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint substitute the linear equation into the other

Question 3

(4 marks)

Solve the simultaneous equations

$y = x^2 - 2x$

$y = x + 4$

.....

Question 4

(4 marks)

Solve the simultaneous equations

$y = x^2 + 1$

$y = 3x - 1$

.....

Exam Practice — Lines and Circles

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint substitute the linear equation into the other

Question 5

(4 marks)

Solve the simultaneous equations

$x^2 + y^2 = 17$

$y = x + 3$

.....

Question 6

(4 marks)

Show that the line $y = x + 6$ does not meet the circle $x^2 + y^2 = 9$.

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

x = −1, y = 0 — B1

x = 2, y = 3 — B1

Tip: give both x and y.

Question 2

x = 3, y = 4 — B1

x = 4, y = 3 — B1

Tip: check: 9 + 16 = 25.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x^2 - 2x = x + 4$ — M1

$x^2 - 3x - 4 = 0$ — M1

$(x - 4)(x + 1) = 0$ — A1

$x = 4, y = 8;\ \ x = -1, y = 3$ — A1

Tip: put both answers into $y = x + 4$.

Question 4

$x^2 + 1 = 3x - 1$ — M1

$x^2 - 3x + 2 = 0$ — M1

$(x - 1)(x - 2) = 0$ — A1

$x = 1, y = 2;\ \ x = 2, y = 5$ — A1

Tip: use the linear equation to find y.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x^2 + (x + 3)^2 = 17$ — M1

$2x^2 + 6x - 8 = 0$ — M1

$(x + 4)(x - 1) = 0$ — A1

$x = 1, y = 4;\ \ x = -4, y = -1$ — A1

Tip: expand $(x + 3)^2$ carefully.

Question 6

$x^2 + (x + 6)^2 = 9$ — M1

$2x^2 + 12x + 27 = 0$ — A1

$b^2 - 4ac = 144 - 216 = -72$ — M1

< 0, so no solutions — A1

Tip: discriminant < 0 means no crossing.